

1 Lesson 24-09-26

We continue treating *active components*, starting from the **current-voltage** characteristic of the Diode:

Definition 1.1: Schottky's law

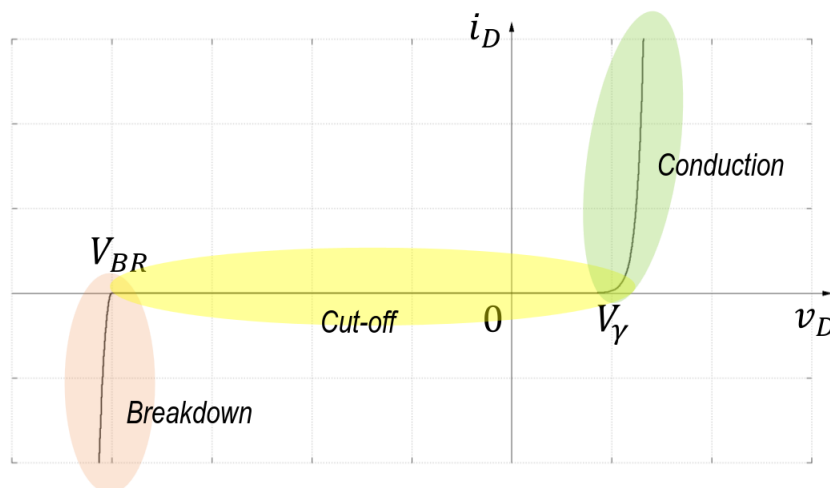
The current-voltage characteristic of a diode is:

$$i_D(t) = I_S \left(e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$$

Where:

- I_S is the *saturation current*, interpreted as the maximum reverse current we can force through the diode;
- η is the *ideality factor*, between 1 and 3 (usually simplified to 1);
- V_T is the *voltage equivalent of temperature*, around 26 millivolts.

Of course, the breakdown effect experienced by the diode for $v_D(t) \ll 0$ isn't taken into account in Schottky's law. We usually take approximated models where the diode starts acting like a voltage source if the reverse current conduction constraint is satisfied.

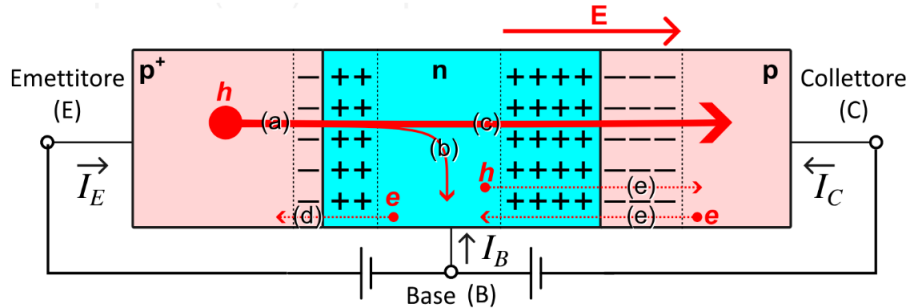


Diodes can be *modeled* in a variety of simplified ways:

- **Ideal switch**: when voltage drop is negative, current is 0; when current is positive, the voltage drop is 0;
- **Ideal switch with offset**: when voltage drop is less than V_γ , the current is 0; when current is positive, the voltage drop is V_γ ;
- **Ideal switch with dynamic resistance**: when voltage drop is less than V_γ , the current is 0; when voltage drop is more than V_γ , the current follows a resistive law. This is equivalent to applying a small series resistance when the ideal diode is conducting.

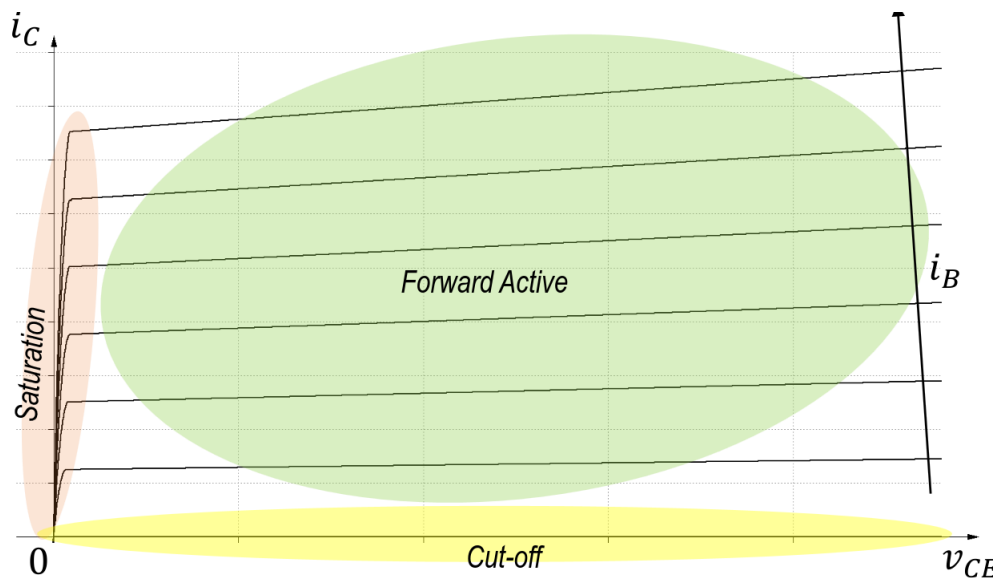
The way circuits are solved by using approximate models is by first making an *assumption* about the state of a diode (conducting or not), and then checking that assumption. Since the models have 2 states, for a diode we can assume that if it's not one state, it's the other. For n diodes we have to make n choices, and thus explore a 2^n assumption space.

BJT transistors are active semiconductor components, commonly used as amplifiers or current controlled switches. They also are a non-linear component, built from 2 PN junctions put head-to-head (which we know on their own made diodes). Two types of BJT transistors exist: PNP and NPN. NPNs are more common.



The BJT transistor is *tripole* made up of:

- **Emitter:** emits charge carriers scaled by the base current;
- **Base,** used to control the device via current;
- **Collector,** collects charge carriers emitted by the emitter.



BJTs work in 3 main regions:

- **Forward active** region, where the collector current is βi_B , so a direct scaling of the base current by a gain factor β . The voltage drop of the base on the emitter is taken with the usual diode law as $V_\gamma \approx 0.7V$. A *reverse active* region also exists but is usually undesirable;
- **Cut-off** region, where the collector current is 0 and the BJT acts as an open switch;
- **Saturation** region, where the v_{CE} between collector and emitter is approximately 0 (around $0.1 \sim 0.2V$). In this region the collector cannot give more current and the BJT acts as a closed switch.

MOS transistors are also active semiconductor components, like BJTs, commonly used as amplifiers or current controlled switches. They use a different technology to

obtain basically the same characteristics: historically they were proposed around the time BJTs started becoming popular, but weren't implemented until much later.

The main difference is that MOS transistors act as *voltage controlled*, instead of *current controlled* devices.

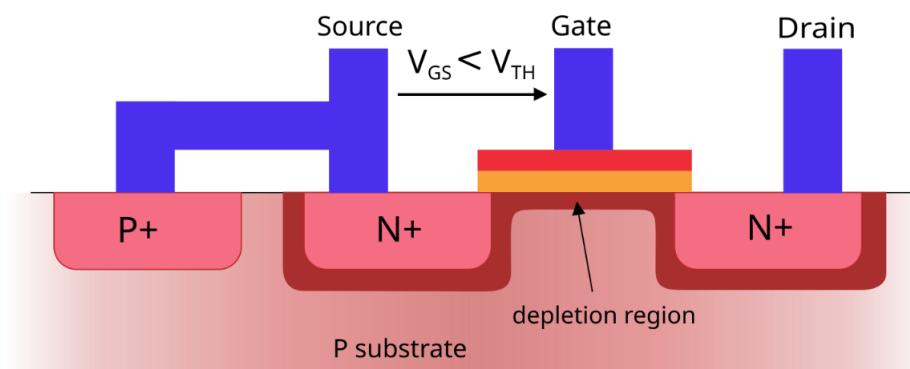
MOSes consist, briefly, of three superimposed layers:

- A base layer of doped semiconductor, which is usually silicon (Si) doped P;
- An intermediate oxide layer, which historically was silicon oxide (SiO_2) and today is mainly polysilicate. In general, we assume that the oxide is a perfect insulator (in reality, it never is, but this approximation is useful to us);
- An upper metal layer, which historically was aluminum (Al).

There are then two additional electrodes, connected to N-doped regions in the P-type substrate. We call these electrodes **source** and **drain**. At this point, the bottom electrode G is called **gate**.

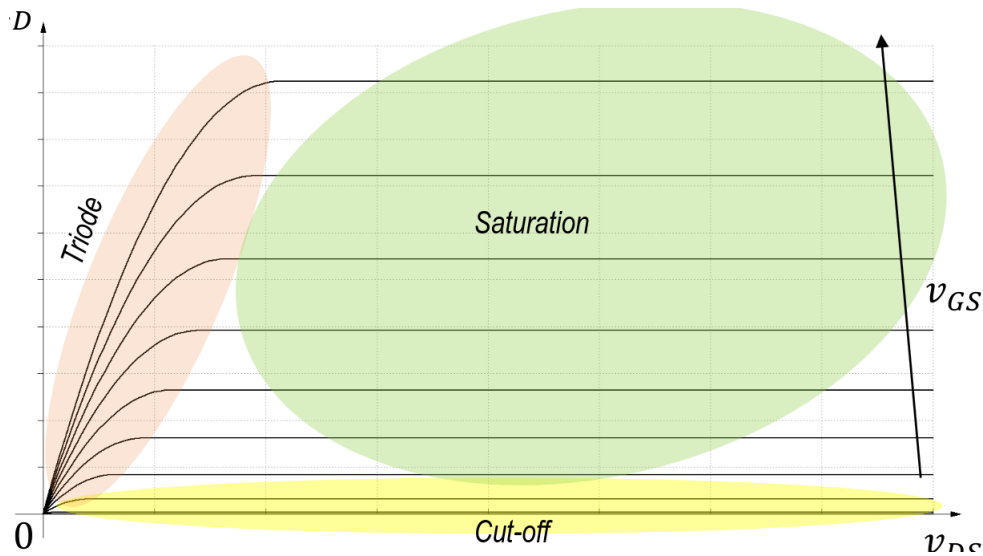
We can see that between the P-type substrate and the two N-type electrodes (source and drain), *diodes* are essentially formed, preventing current from flowing from the source and drain into the substrate.

For the MOSFET to operate, therefore, the N-doped regions need to extend beneath (or at least slightly overlap) the oxide. We call the distance L between the two N-doped regions the *gate length*.



So, we can say that a MOS transistor is another *tripole* made up of:

- **Source**: emits charge carriers scaled by the base voltage;
- **Gate**, used to control the device via voltage, it is always isolated from drain and source;
- **Drain**, collects charge carriers emitted by the emitter.



MOSes work in 3 main regions:

- **Cut-off** region: when the v_{GS} voltage between gate and source is lower than a certain v_{th} threshold voltage, the drain current is zero and the MOS behaves as an open circuit;
- **Triode** or *linear* region: when the v_{GS} voltage is higher than v_{th} , but v_{DS} is less than $v_{GS} - v_{th}$, that is:

$$v_{GS} > v_{th}, \quad v_{DS} < v_{GS} - v_{th}$$

then we are in a linear region where current scales mostly directly with voltage applied to v_{DS} . The exact law is:

$$i_D = \mu C_{ox} \frac{W}{L} \left((v_{GS} - v_{th})v_{DS} - \frac{v_{DS}^2}{2} \right)$$

where C_{ox} is the oxidation coefficient of the oxide (usually a polysilicate), and W and L are respectively the width and the length of the MOS gate channel;

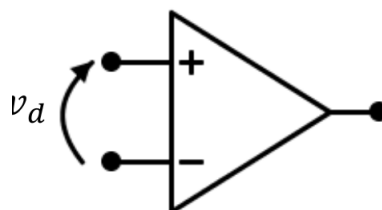
- **Saturation** region: when the v_{GS} voltage is higher than v_{th} , but v_{DS} is higher than $v_{GS} - v_{th}$, that is:

$$v_{GS} > v_{th}, \quad v_{DS} > v_{GS} - v_{th}$$

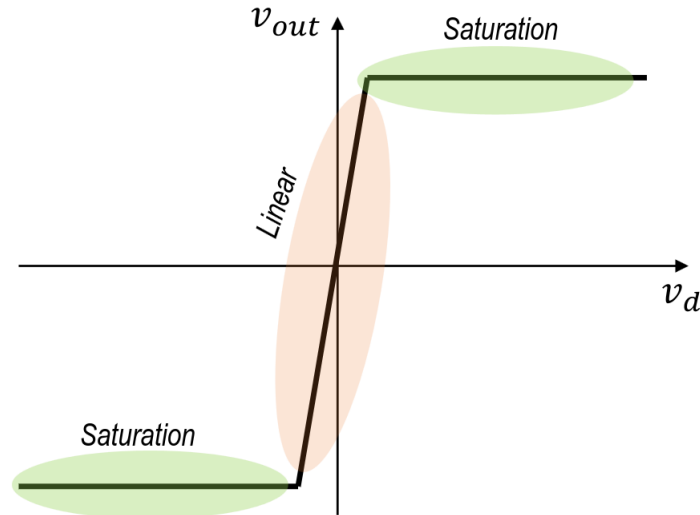
then we are in saturation, where the current scales directly with v_{GS} independently of v_{DS} . The exact law is:

$$i_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})^2$$

Operational amplifiers are complex components comprised of DC-coupled differential amplifiers with very high gain values.



What OP amps basically do is take the difference in voltage between two terminals (usually labeled + and -) and amplify that difference with an extremely high open loop gain (G_{OL}).



This means the output characteristic is:

$$v_{out} = G_{OL}(v^+ - v^-) = G_{OL}v_D$$

Let us note that, even if G_{OL} is in the order of several millions, an OP amp is a powered device which requires power and ground rails, and that cannot return voltages lower than the ground rail or higher than the power rail (a phenomenon known as *saturation*).

They are most commonly used as a **linear amplifier**:

- With a negative feedback network;
- Forced to work in linear region;
- Lower final gain, larger bandwidth, stability.

Or used as a **comparator**:

- With a positive feedback network (or no feedback);
- Forced to work in saturation regions.

They are widely used in many embedded systems.

Below, a few common circuits with OP amps are noted.

- Non inverting amplifier

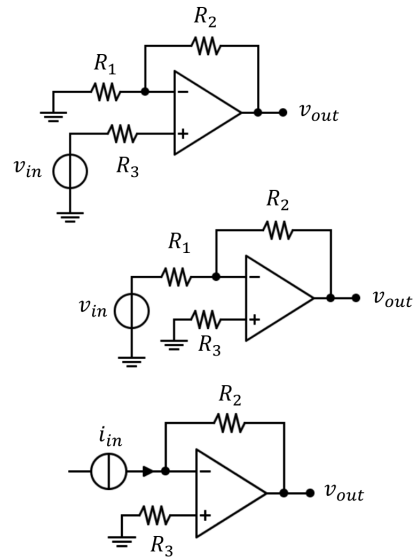
- $v_{out} = v_{in} \left(1 + \frac{R_2}{R_1}\right)$, $R_{in} = \infty$, $R_{out} = 0$
- R_3 used to reduce unwanted output offsets

- Inverting amplifier

- $v_{out} = v_{in} \left(-\frac{R_2}{R_1}\right)$, $R_{in} = R_1$, $R_{out} = 0$
- Note the non ideal (low) input resistance

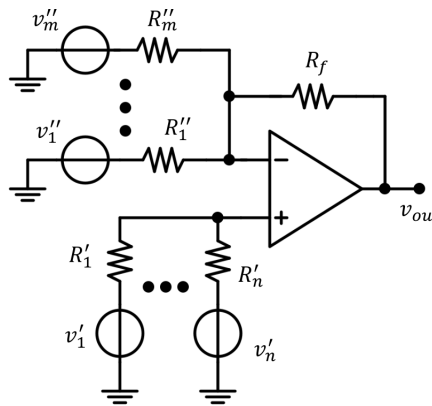
- Transresistance amplifier

- $v_{out} = -R_2 i_{in}$, $R_{in} = 0$, $R_{out} = 0$



We note that transresistance amplifiers will be useful to us when we need to implement analog converters.

Generalized analog adders / subtractors can be designed via OP amps by connecting inputs we want to add to the non inverting input and inputs we want to subtract to the inverting input.



Let's make sure to clearly define what we're after before delving into the implementation details. We ideally want a network based on a negative feedback OP amp amplifier, with a single R_f resistance as the feedback network.

- On the non-inverting input, we'll connect in parallel n voltage generators $v'_1, v'_2, \dots, v'_n$, with series resistances of $R'_1, R'_2, \dots, R'_n$;
- On the inverting input, we'll connect in parallel m voltage generators $v''_1, v''_2, \dots, v''_m$, with series resistances of $R''_1, R''_2, \dots, R''_m$.

We want the output of this network to perform the following operation on the voltage sources, taken as inputs:

$$V_{out} = \sum_{i=1}^n A'_i v'_i - \sum_{i=1}^m A''_i v''_i$$

- Let's recall that the inverting input gain can be taken as:

$$V_{out_j}'' = -\frac{R_f}{R_j''} v_j''$$

- Whereas the non-inverting input gain can be taken as:

$$V_{out_j}' = \left(1 + \frac{R_f}{\prod_{i=1}^m R_i''}\right) V_j'$$

The idea is to choose the resistances R_i' and R_i'' such that the expression of V_{out} above holds.

- Choosing the resistances on the inverting side is really easy as we can simply say:

$$R_i'' = \frac{R_f}{A_i'}$$

and in the inverting input gain equation we get the correct gain;

- For the inverting side, we first compute the voltage on node V^+ through Millman's theorem (theorem 1.2, of which we gave a proof in section 1.2.5):

$$V^+ = \frac{\sum_{i=1}^n \frac{v_i'}{R_i'}}{\sum_{i=1}^n \frac{1}{R_i'}}$$

From this equation we see that a good starting point is setting:

$$R_i' = \frac{R_f}{A_i''}$$

so that the gains at the numerator are correct:

$$V^+ = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'}$$

Now we use the non-inverting input gain to compute the output of the non-inverting side:

$$\begin{aligned} V_{out}' &= V^+ \left(1 + \frac{R_f}{\prod_{i=1}^m R_i''}\right) = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \frac{R_f}{\prod_{i=1}^m \frac{1}{R_i''}}\right) \\ &= \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + R_f \sum_{i=1}^m \frac{1}{R_i''}\right) = \frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \sum_{i=1}^m A_i''\right) \end{aligned}$$

At this point we impose the output be equal to the positive side of the original output equation:

$$\frac{\sum_{i=1}^n A_i' v_i'}{\sum_{i=1}^n A_i'} \left(1 + \sum_{i=1}^m A_i''\right) = \sum_{i=1}^n A_i' v_i' \implies \frac{1 + \sum_{i=1}^m A_i''}{\sum_{i=1}^n A_i'} = 1$$

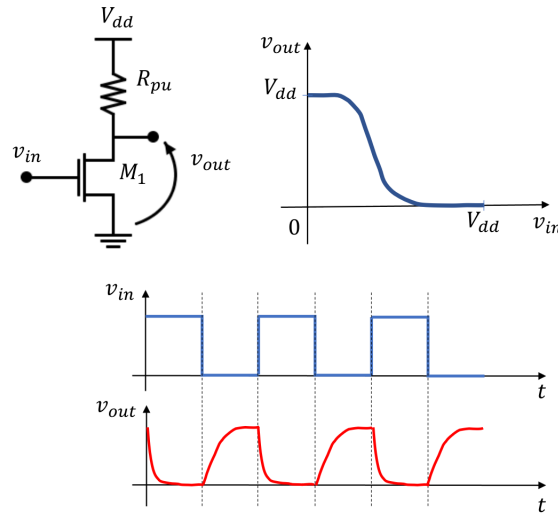
This last equation we obtained is what we have to satisfy in order to get the correct resistor values. It's pretty simple to make the result right by just adding a resistance on the inverting or non-inverting path based on the value of the left hand side, which we'll call α :

$$\alpha = \frac{1 + \sum_{i=1}^m A_i''}{\sum_{i=1}^n A_i'}, \quad \begin{cases} A_{n+1}' = 1 - \sum_{i=1}^n A_i' + \sum_{i=1}^m A_i'' \\ A_{m+1}'' = -1 + \sum_{i=1}^n A_i' - \sum_{i=1}^m A_i'' \end{cases}$$

1.1 Logic gates

We can move on to digital electronics concept, by introducing **logic gates**. A logic gate is an electronic component which performs some sort of operation on digital signals.

A simple example of a logic gate is an **inverter**. The simplest inverter is given by a *resistive load* inverter.



What happens in the circuit is that if we raise v_{in} , and thus v_{GS} , above the threshold voltage v_{th} , then we get conduction between the drain and source of the MOS transistors, and the circuit acts as an open switch. If we then sample v_{out} , we see the voltage after the fall on R_{pu} (which will be close to zero).

In the other two graphs we see two non idealities of this circuit:

- **Asymmetric** response: the falling edge is much sharper than rising edge, as the R_{DS} of the NMOS is much smaller than R_{pu} ;
- We have **static** power consumption in the on state (when output is at logic 0), as current flows from drain to source:

$$P = \frac{v_{DD}^2}{R_{DS} + R_{pu}}$$

- We also have a certain amount of **dynamic** power consumption, given by the transients happening at state changes. These rely on equivalent capacitances of wires and MOS transistors, and will be studied in more detail later.

We can then see that we need a better design, at least in term of response alignment, and absolutely without static power consumption.